Niche-Conditioned Regulatory Transport in Premalignant Epithelial Progression

A reproducible context-residual estimand and the spatial scale of its recovery

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The paradox that defines premalignancy

concept schematic

a lesion field —
many indolent, one progressing

Same mutations.

Same histology.

Different outcomes.

Two explanations

Cell-intrinsic

the tissue is a bystander

Tissue-held

the tissue is a participant

Holding a cell's own state and stage fixed,
does its tissue context change where it ends up?

The premalignant niche: a candidate mechanism

FIGURE NOT FOUND

biorender_mechanism

The measurement will not cooperate

Profiling destroys the tissue.

No cell is ever observed twice.

So the effect cannot be measured. It has to be *estimated*.

Which means the design has to be built around falsification, not fit.

Three hypotheses, fixed before the data

H_1 A reproducible context residual exists.

H_2 That information is *receiver-local*.

H_3 The estimand transfers to a second epithelium.

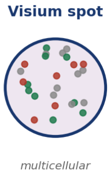
They do not resolve in the same direction.

That is the contribution.

Act I

The estimand is identifiable, and it is not circular

One measurement, two disjoint readings



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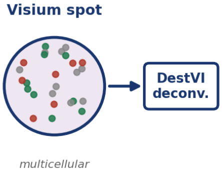

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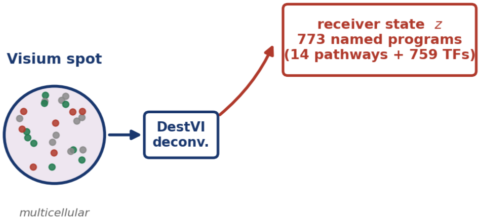

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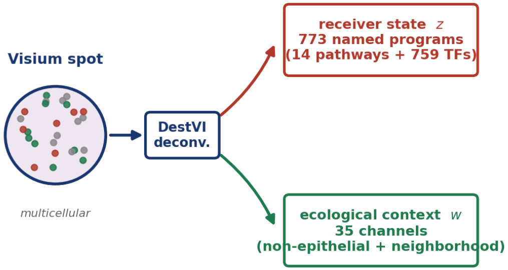

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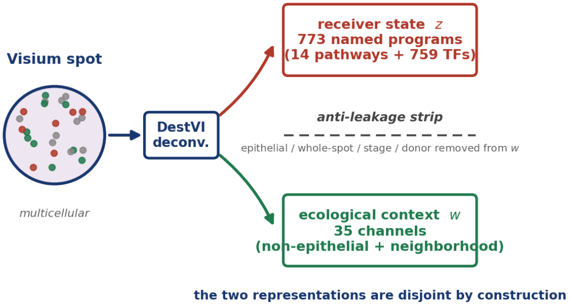
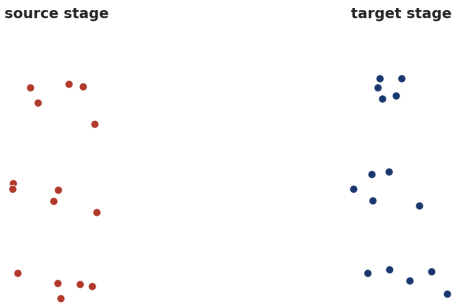


Correspondence is estimated *within* ecological strata



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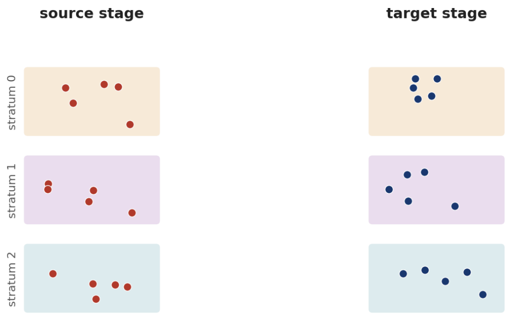

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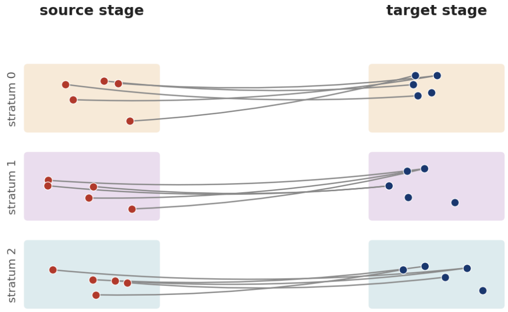

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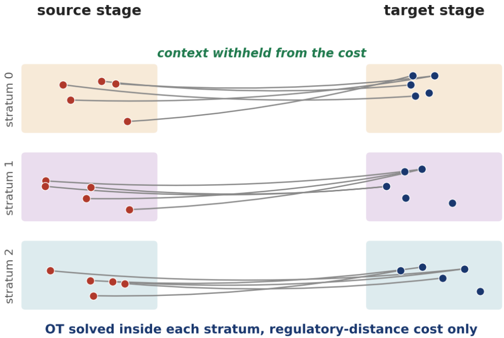

The field is structured, and context enters through a waist

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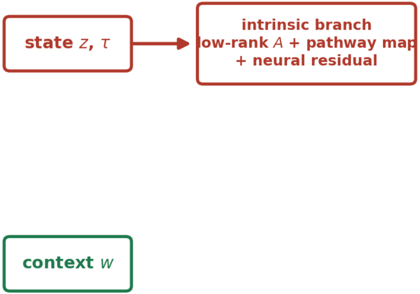

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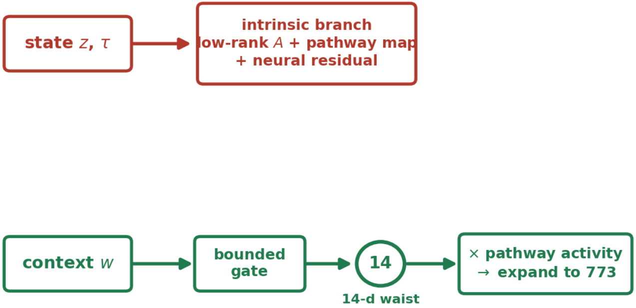

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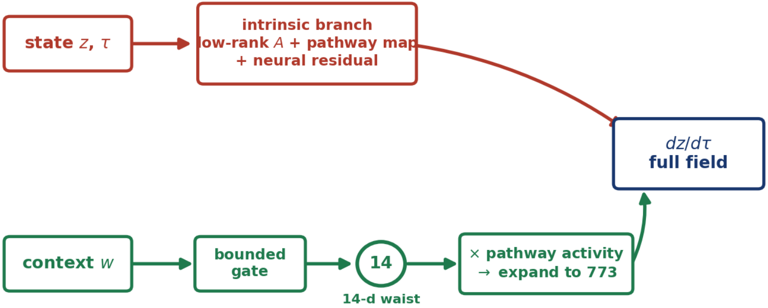

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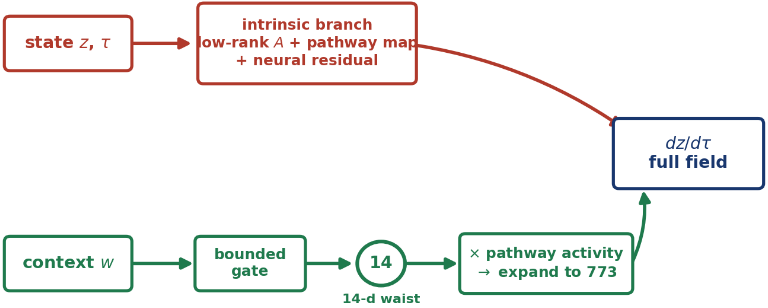

759 TF coordinates move only through the 14-channel gate or the residual

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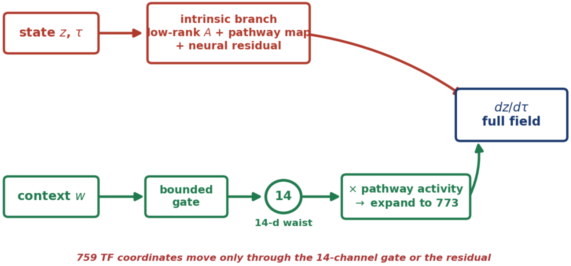
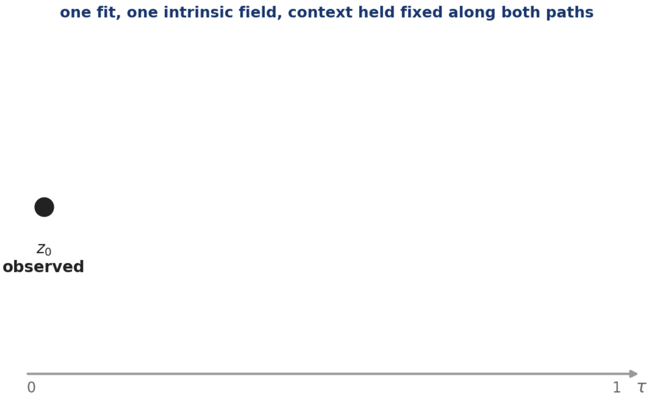


The learned field, made explicit:

$$\frac{dz}{d\tau} = \underbrace{f_{\text{self}}(z,\tau)}_{\text{intrinsic}} + \underbrace{g(w) \odot f_{\text{ctx}}(z,\tau)}_{\text{niche-gated, 14-d gate } g}$$

One start state, two integrations



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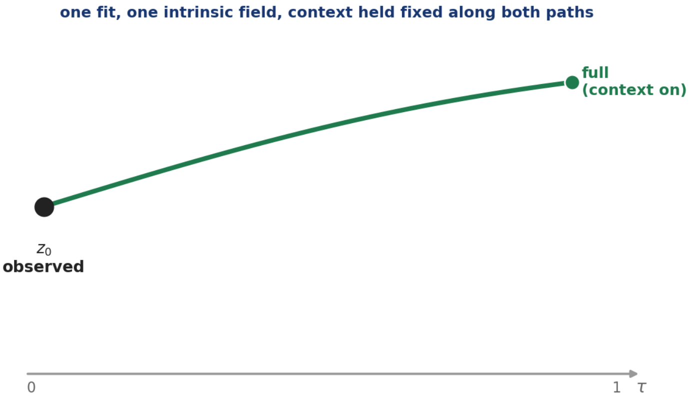

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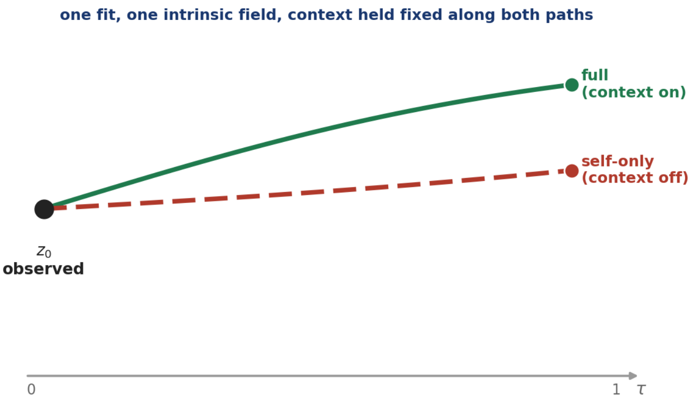

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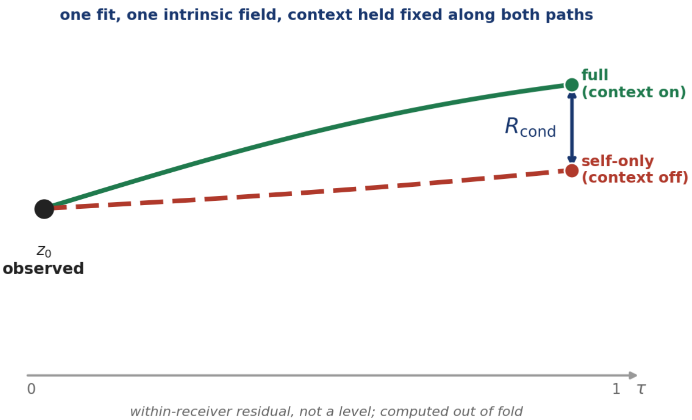

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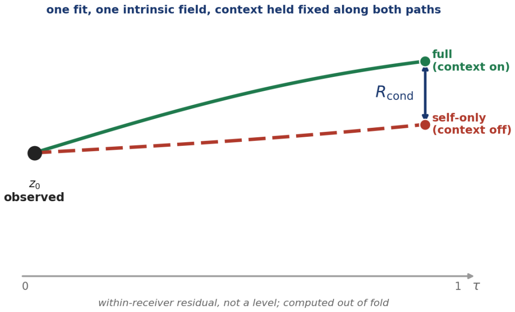
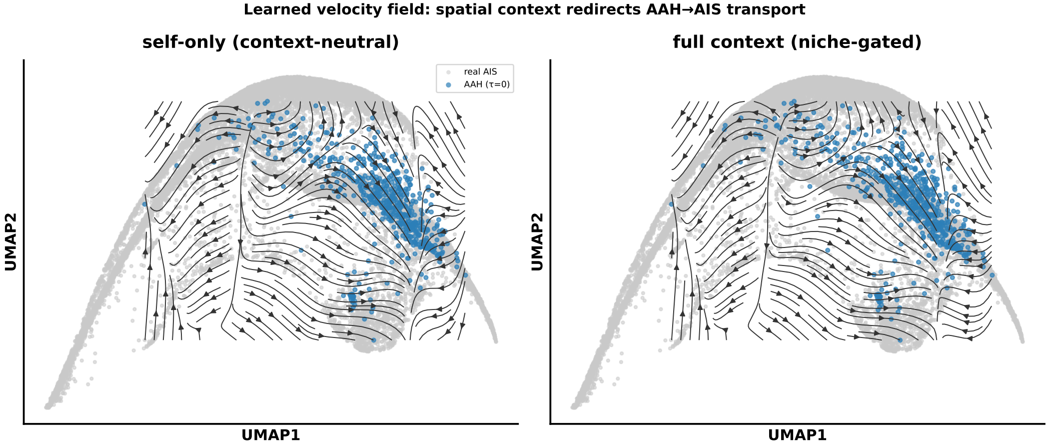


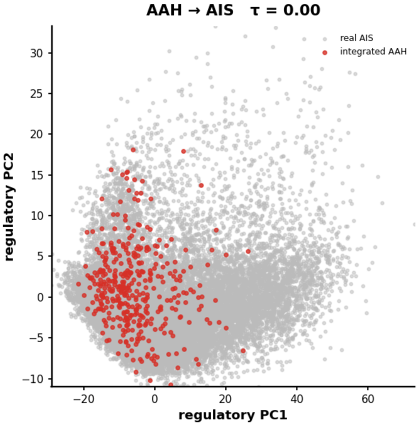
The estimand, one line:

$$R_{\text{cond}}\left(z_i^{\text{obs}}, w_i\right) = z_i^{\text{full}}(1) - z_i^{\text{self}}(1) \qquad \text{(same } z_0\text{, same intrinsic field, context on vs. off)}$$

The fitted field, both branches, over the real manifold



Watching the transport: integrated AAH lands on real AIS



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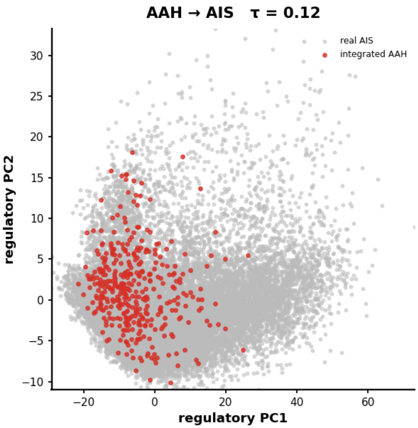

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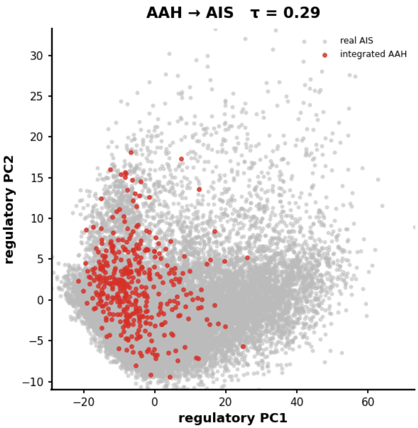

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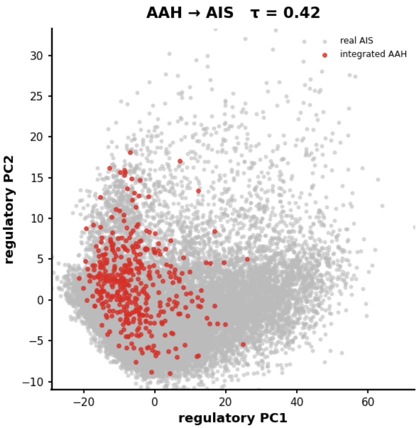

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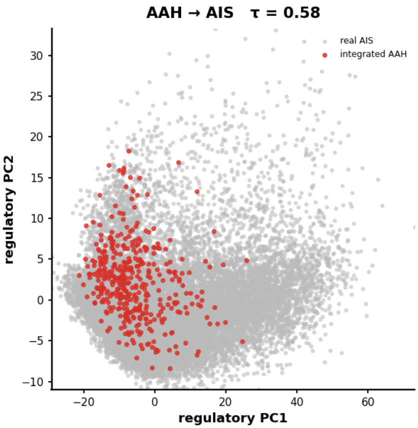

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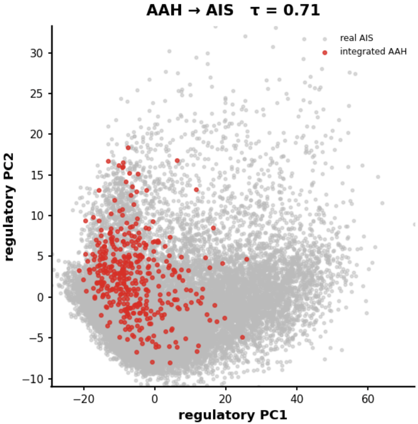

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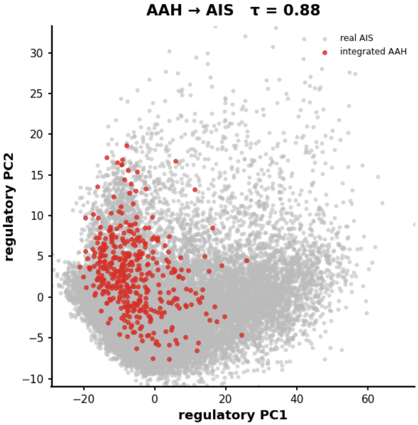

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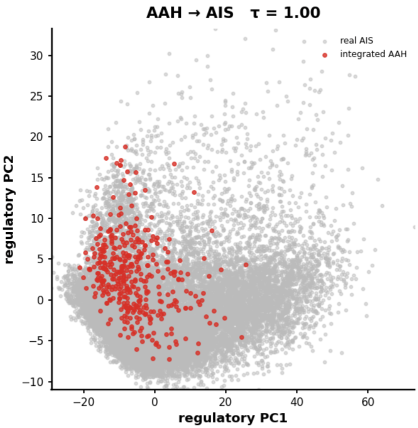


My first design failed its own screen

Three candidate estimands.

Two were rejected.

One of them was mine.

The context vector does not encode the receiver

ridge regression, ecological context $\rightarrow$ receiver state, held-out

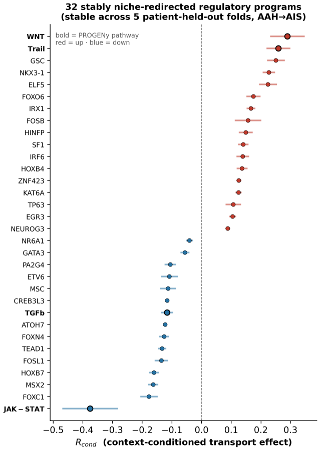
$R^2 < 0$ in all five folds

mean -0.54 worse than predicting the mean

Act II

What the estimand finds

29 of 773 programs are reproducibly niche-redirected



The preinvasive residual is a circuit, not a list

Down

ETV3 ETV6
ZBTB18 GLI3

four transcriptional repressors

Up

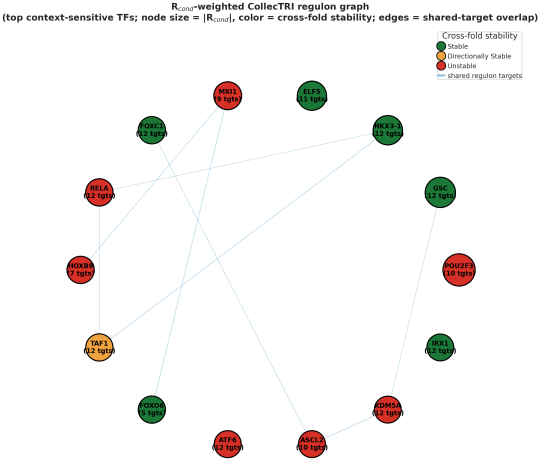
FOS

p53 RB1 KAT5

their target, and the checkpoint

Loss of repression under retained restraint.

The preinvasive residual is a circuit, not a list

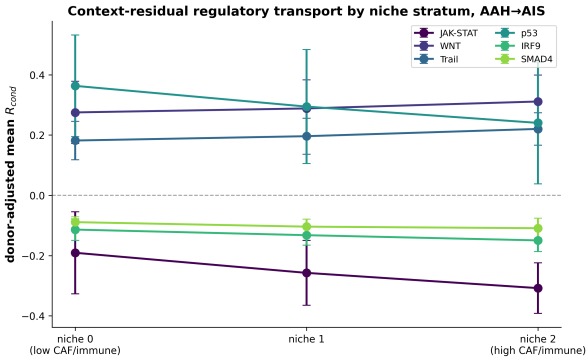


Same configuration. Different organ. Different method.

This work human LUAD precursors context-residual transport	**Reyes et al., Cell 2026** mouse PDAC, spontaneous p53 loss single-cell + spatial lineage
p53 ↑ RB1 ↑ SMAD4 ↓, with WNT / EGFR / AP-1 ↑	p53, CDKN2A, SMAD4 **co-activated** with oncogenic programs in progenitor-like cells

Including the same arm — SMAD4 — giving way.

The niche grades the contest



Less restraint, more drive at the inflamed end.

Interferon tone is lost *before* invasion

	AAH $\rightarrow$ AIS	AIS $\rightarrow$ invasive
JAK-STAT	-0.246	—
IRF9	-0.129	—
IFN-γ response	-1.85	$+1.78$
IFN-α response	-1.75	$+1.80$

two readouts, different confounds, same direction

And in lesions with known outcomes, that loss predicts progression

Progressive lesions: interferon lost

Regressive lesions: interferon retained

the only human premalignancy cohort profiled *and* followed to a known fate

Invasion is a change of architecture, not more drive

Down

HDAC1 CTBP1 NAB2

three corepressors

Up

MAML1 FOXA1

coactivator and pioneer factor

Lineage identity is dismantled here, not before.

An anabolic-stress module, on the same edge

Context residual

ATF5 HSF2 NFKB2

stress response and proteostasis

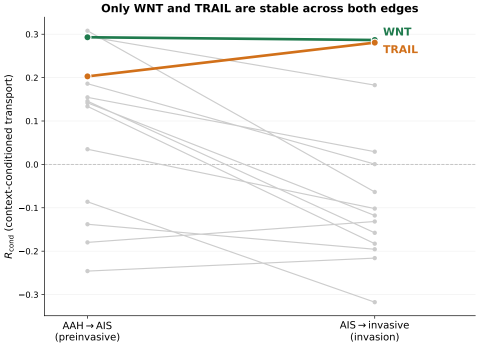
Whole transcriptome

MYC targets OxPhos

both reversed from preinvasive

Anabolic demand and its compensation, together.

Only two programs are invariant across the boundary



WNT **TRAIL**

everything else is edge-specific

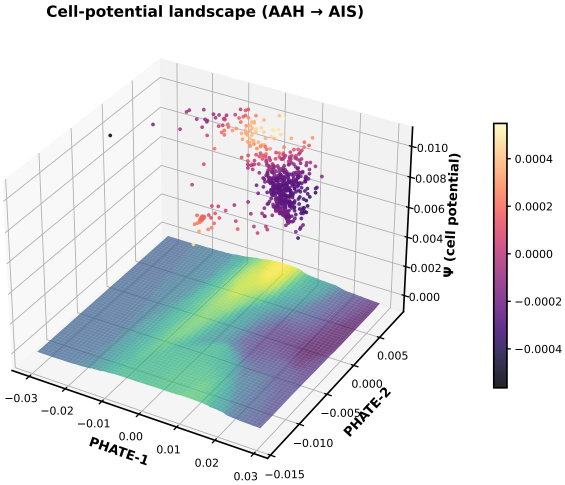
VEGF goes *down* at invasion — and that is consistent

$$\text{VEGF} \qquad R_{\mathrm{cond}} = -0.318$$

largest stable pathway at the invasion boundary

A residual is not a level.

Progression is a directed flow, not a relaxation



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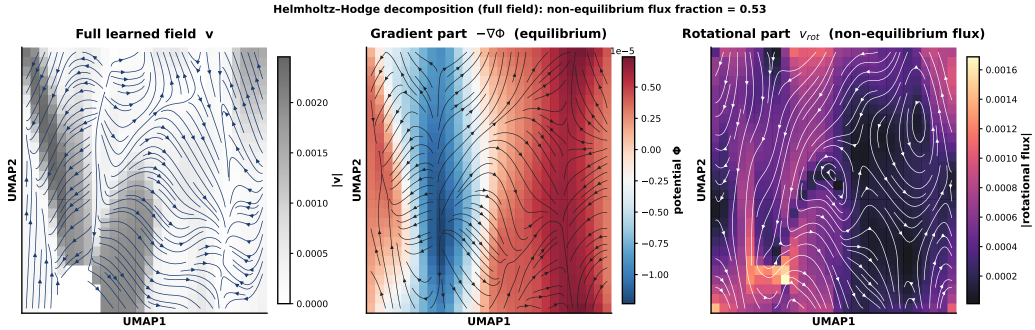


The gate recovers the return arm of a known circuit

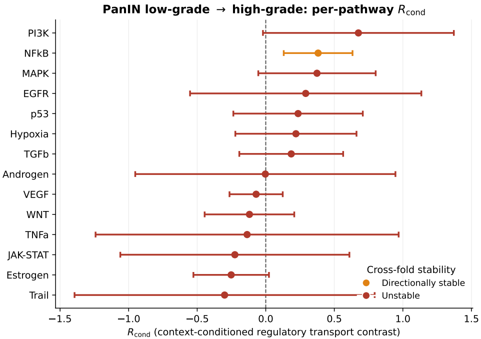
$\sim$1.3 million candidate signalling chains

$\downarrow$

two programs

macrophage APOE, PSAP, A2M $\rightarrow$ LRP1, EGFR $\rightarrow$ AP-1

The estimand transfers without over-calling



Point: across-fold mean. Whiskers: ±1 SD across 3 patient-held-out folds (4 donors).

Four donors. **Zero stable programs.**

Act III

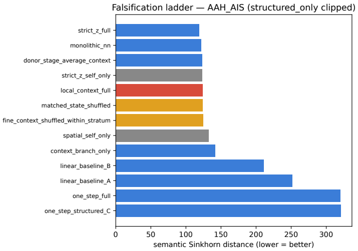
At what spatial scale does this hold?

The falsification ladder

Rung	What it removes
Local context	nothing — reference
Shuffled context	**niche identity, distribution kept**
Donor–stage average	within-specimen variation
Self-only	context entirely
Strict-state coupling	the ecological stratification

paired across five patient-held-out folds

A split verdict — and one half is a bound



Paired contrast	Δ	95% CI
Local over **shuffled**	+0.08	[−1.10, +1.26]
Local over **self-only**	+8.76	[−2.83, +20.35]

Why one interval is tight and the other is not

	F0	F1	F2	F3	F4	SD
self − local	+5.82	+22.04	+14.26	+2.73	−1.06	9.33
local − shuffle			*folds co-move*			≈ 1.0

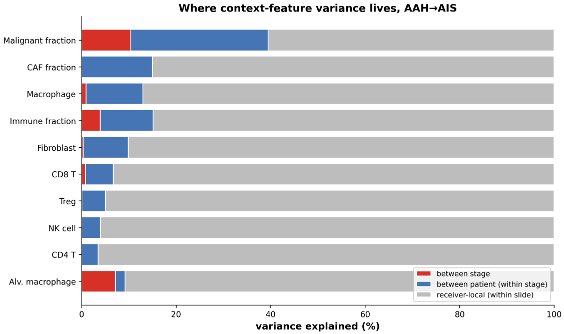
Structural, not arbitrary.

At the invasion boundary, both comparisons are null

Paired contrast	Δ	95% CI
Local over shuffled	$+0.28$	$[-5.82, +6.38]$
Local over self-only	-4.59	$[-13.30, +4.11]$

zero shared donors on this edge

Why: the ecology varies below our resolution



85–96% receiver-local and stage-invariant

between-patient 8–15% between-stage <1%

An unrelated modality measured that scale

Immune hot and cold spots interchange over

tens of microns

Our spots are **55 µm** at **100 µm** centres

Two readings, and the experiment that separates them

A. Sub-resolution

local, but below 100 μm

better assays recover it

B. Specimen-scale

a field-level property

resolution will not help

Run the identical ladder at single-cell resolution.

A predicts recovery. B predicts continued equivalence.

Act IV

How the hypotheses resolve

Scorecard

	Hypothesis	Verdict
H_1	Reproducible context residual	**Met**
H_2	Receiver-local information	**Not met**
	vs shuffle	bounded equivalence
	vs self-only	underpowered
H_3	Transfers across systems	**Behaviourally**

Weaker than the pre-registered alternative. I am reporting the weaker outcome.

Limitations

- Cross-sectional — couplings are model alignments, not lineage maps
- **Two shared donors** on the primary edge; zero at invasion
- Spot resolution; ecology varies at or below it
- Context read at source and held fixed — no niche co-evolution
- 759 factors move through a **14-dimensional** gate
- Co-expression and communication are cross-method, not independent
- **Retrained-shuffle null: specified, not yet run**

What comes next

1. Retrained-shuffle null — the outstanding confirmatory test
2. **The identical ladder at single-cell resolution**
3. Global/local decomposition, if ecology is field-scale
4. A mechanistic context channel: receptor occupancy, not a learned gate
5. Perturb APOE/PSAP/A2M $\rightarrow$ LRP1/EGFR $\rightarrow$ AP-1

Each specified precisely enough for someone else to run.

Contribution

A falsifiable estimand.

A screen that rejected my own first design.

Reproducible, named biology on two stage edges.

A measured bound on where it stops.

Chris Bradburne

committee, collaborators, compute

Questions

Backup

B1 — Is 29/773 just the false-positive rate?

- $29/773 = 3.8\%$; a single 95% CI criterion has a 5% naive null rate. Fair question.
- Criterion is not one test: CI excluding zero **and** sign-consistency across five *disjoint patient* folds.
- Folds are correlated, so I do not claim an exact null rate.
- **The retrained-shuffle null is the right test and is not yet run.**
- Cheap intermediate: permute strata within donor, re-run the filter, report the null stable-count distribution.

B2 — The context channel is 14-dimensional

- Context enters only via $\mathbb{R}^{35} \to \mathbb{R}^{14} \to \mathbb{R}^{773}$.
- So coordination among transcription factors is partly *expected*, not evidential.
- What the bottleneck does *not* determine is which factors load or with what sign. A rank-restricted map does not force four repressors to oppose their target. **Sign structure is the informative part.**
- Pathway-level claims sit directly in the gated block and are strongest.

B3 — Per-fold ladder, preinvasive edge

	F0	F1	F2	F3	F4	mean
Local context	103.32	100.12	113.93	149.47	151.42	123.65
Self-only	109.14	122.16	128.19	152.20	150.36	132.41
self − local	+5.82	+22.04	+14.26	+2.73	−1.06	+8.76

SD 9.33, SEM 4.17, $t = 2.10$, $P \approx 0.10$. Against shuffle, SD ≈ 1.0.

B4 — Decomposition credibility per fold

	F0	F1	F2	F3	F4	mean
D_ρ	0.949	0.963	1.008	0.984	0.922	0.965

- $D_\rho \leq 1$ means structured terms carry at least as much velocity as the neural residual — the interpretability precondition.
- **Fold 2 fails it.** Reported, not hidden.
- Which is why factor-level claims are stated below pathway-level ones.

B5 — Why is the held-out AUC *below* chance?

- Donor-held-out AUC for the self-only stage classifier: **0.419**, about 1.6 SE below 0.5.
- Not a bug — the signature of a **donor-entangled representation**. Cross-donor generalisation is anti-correlated, not absent.
- **Within-donor AUC is 0.833**, above chance in every shared donor.
- The representation is stage-discriminative; cross-patient transfer is what fails. Which is exactly why every estimate is patient-held-out.

B6 — Baselines and ablations

- Linear endpoint baselines: 210.83 and 251.86 vs 123.65 fitted.
- Single-branch fields collapse: structured-only 2621.22, residual-only 654.81.
- Monolithic network fits *better* (121.59) — the price of an interpretable, program-resolved field. Stated, not omitted.
- Strict-state fits best (119.17) because the metric is scored in the space that coupling optimises. Near-circular, hence rejected on Tier 2.

B7 — Endpoint metric mechanics

- Conditional metric: Sinkhorn cost within held-out strata (≥ 5 per side), source-mass weighted, $\varepsilon = 0.05$.
- Primary because it evaluates the same conditioning that *defines* the estimand.
- Comparisons paired within fold, so estimator bias common across rungs cancels.
- An orthogonal metric — donor-held-out stage classification from predicted endpoints — would harden the negative. Named as future work.

B8 — PanIN power

- Four donors, $\sim$100 graded receivers, two strata, three folds.
- Zero stable; 188 directionally stable.
- **True null and absence of power are not separable** without a positive control at matched n.
- The experiment that would earn the claim: inject a synthetic effect of known magnitude at $n = 4$ and show recovery.

B9 — Provenance

- Every inline number traces to a versioned artifact; audit trail in the Ch3 header and Appendix C.15.
- Folds, hyperparameters, seeds: Appendix C.5–C.7.
- Per-fold ladders, 773-program stability atlas, transport diagnostics: Appendix C.8–C.11.
- Seeded per fold; training donors only; code and environment archived.

Any number on any slide can be traced live.

B10 — Secondary analyses and their evidential status

Tier	Analyses
Model-independent	stage DE, gene-set enrichment, stage discrimination
Residual-associated	ecological localisation, spatial autocorrelation
Residual-guided	SCENIC / hdWGCNA, gated communication

- Only the first tier is independent of the estimand.
- SCENIC overlap is **not enriched** beyond hypergeometric expectation (0.95–1.14$\times$, $P = 0.35$–0.63). Reported as such.

B11 — The ecology is spatially structured

- Moran's I, $k = 15$ per section, 55 sections, 787,816 spots. All 330 section-by-feature tests significant.
- Malignant clustering rises monotonically: $0.234 \rightarrow 0.458 \rightarrow 0.701$; Spearman $\rho = 0.673$.
- Getis–Ord hot-spot fraction nearly doubles.
- But macrophage autocorrelation is high and **stage-flat** ($0.443 \rightarrow 0.499 \rightarrow 0.467$).

Spatially organised within every specimen, only weakly graded by stage — the regime where local niche is real but hard to separate.

B12 — The estimand selection screen

- Frozen before real data: recovery cosine ≥ 0.80, seed agreement ≥ 0.90, null gain < 0.10, non-identifiable correlation < 0.50.
- Five regimes including a confounded null and an alignment sweep.
- Conditional-stratified passed both tiers. Strict-state passed tier 1, failed tier 2. Progression-geometry failed both.
- Oracle pairing was the ceiling and is unavailable to any real analysis.

Two of three candidates rejected, including the one I started with.